

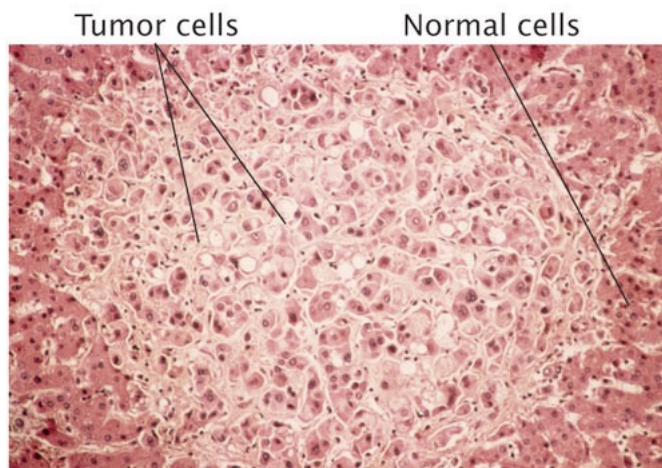
Cancer genetics I

Metastatic breast cancer masses growing in a human liver

(a)



(b)

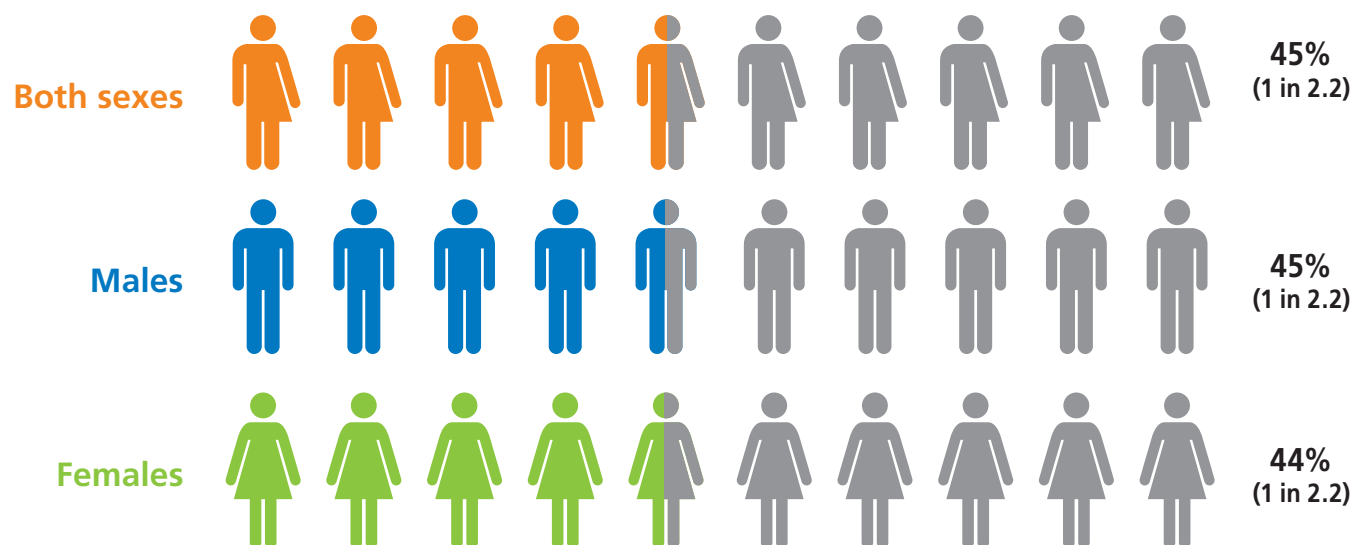


[CNRI/Science Source.]

Cancer statistics in Canada

It is estimated that an average of 675 people in Canada would be diagnosed with cancer and 241 people would die from cancer each day in 2024.

FIGURE 1.1 Lifetime probability of developing cancer, Canada (excluding Quebec and Nova Scotia*), 2019



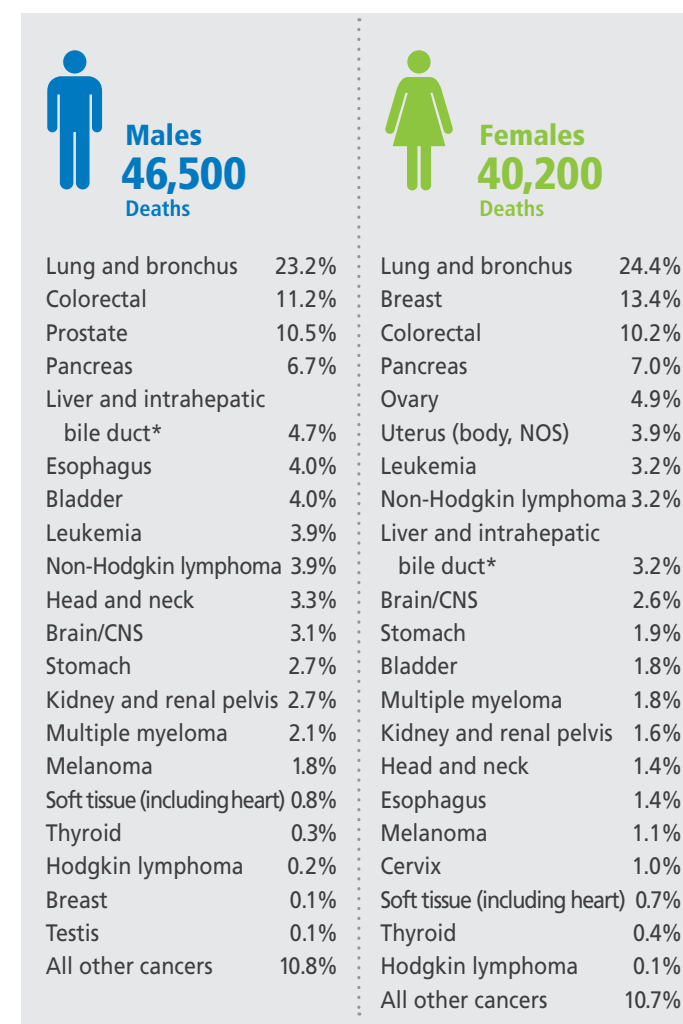
Cancer statistics in Canada

Lung cancer is responsible
for 1 in 4 cancer deaths
in Canada.

FIGURE 2.1 Lifetime probability of dying from cancer, Canada (excluding Quebec and Nova Scotia*), 2020



FIGURE 2.2 Percent distribution of projected cancer deaths, by sex, Canada, 2023



Cancer statistics in Canada

1988 → 2017

Cancer death rates decreased

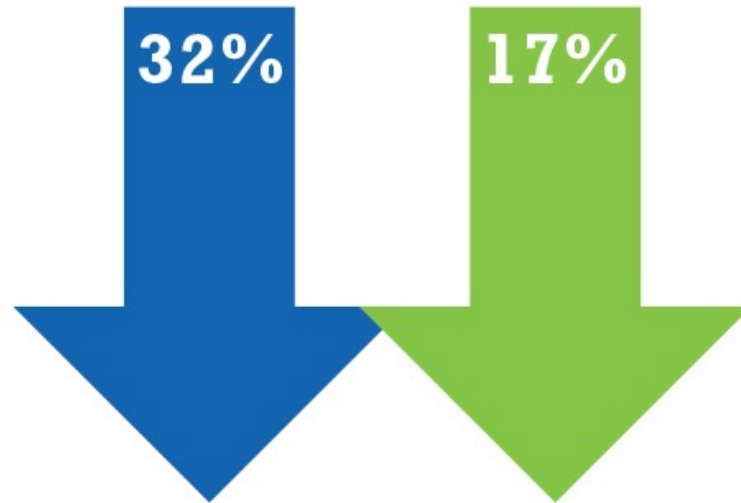
for

males

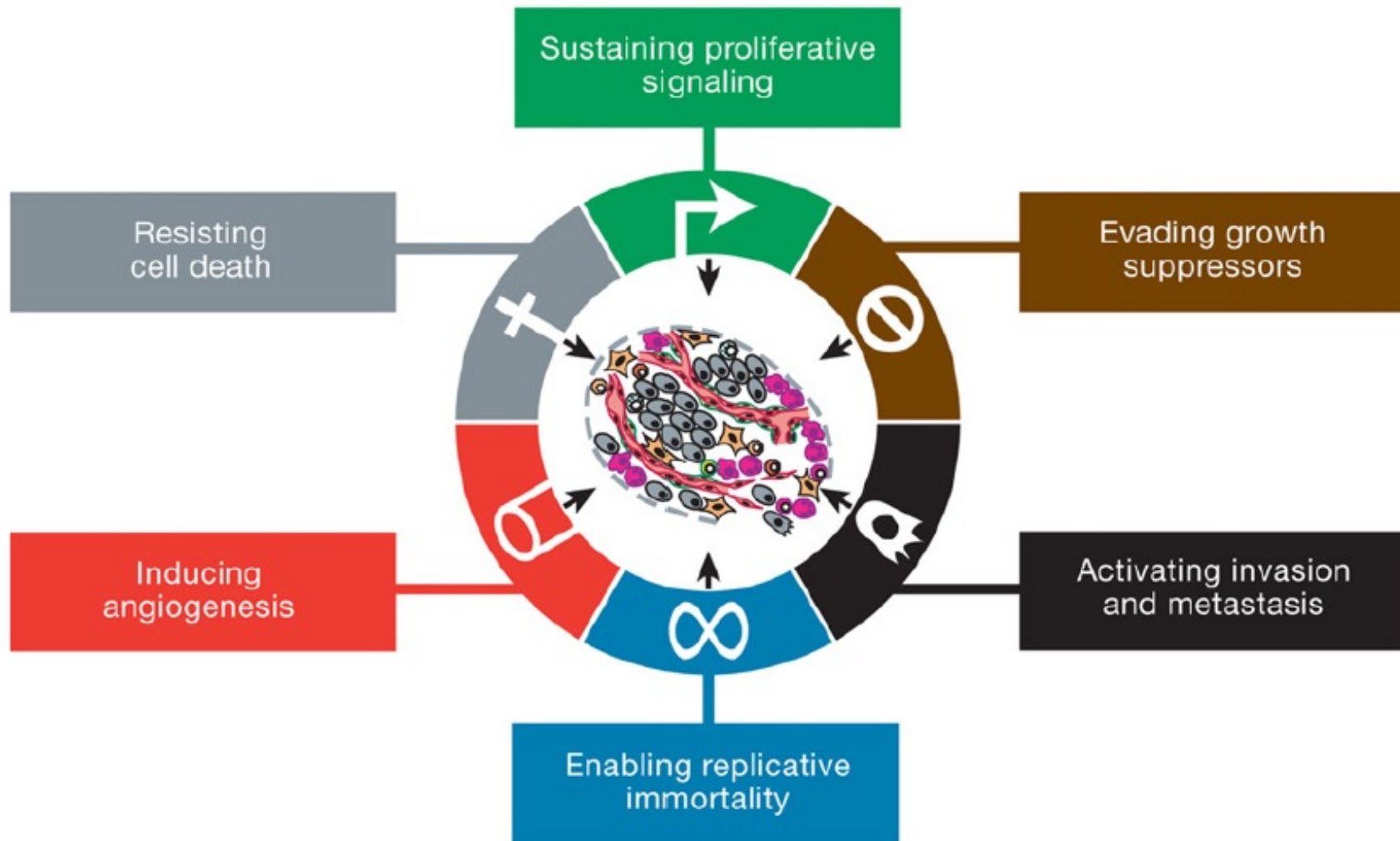
females

32%

17%

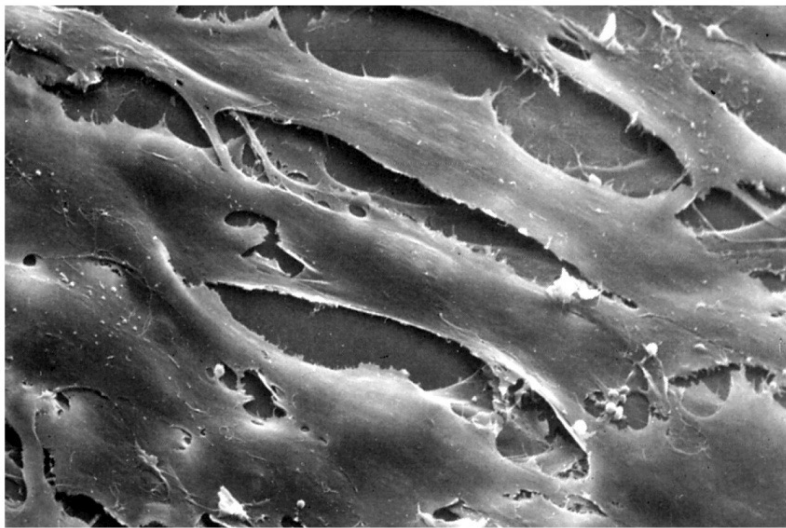


Hallmarks of Cancer I

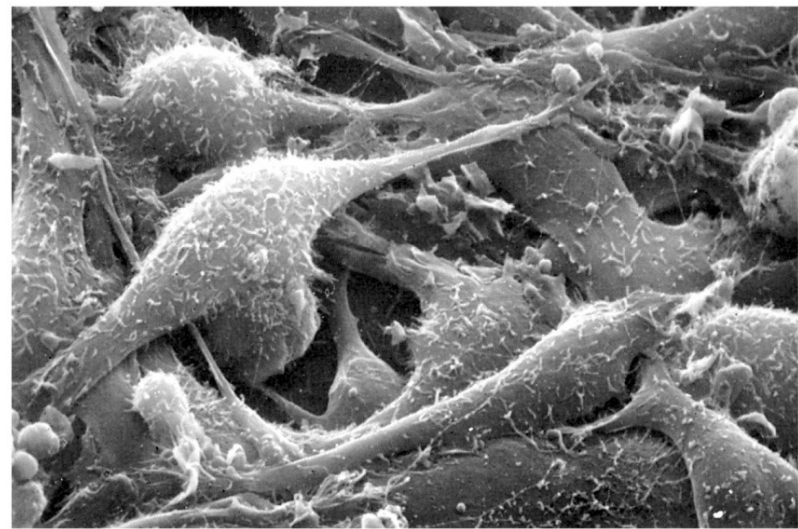


Sustaining proliferative signals/Evading growth inhibition

Normal 3T3 cells

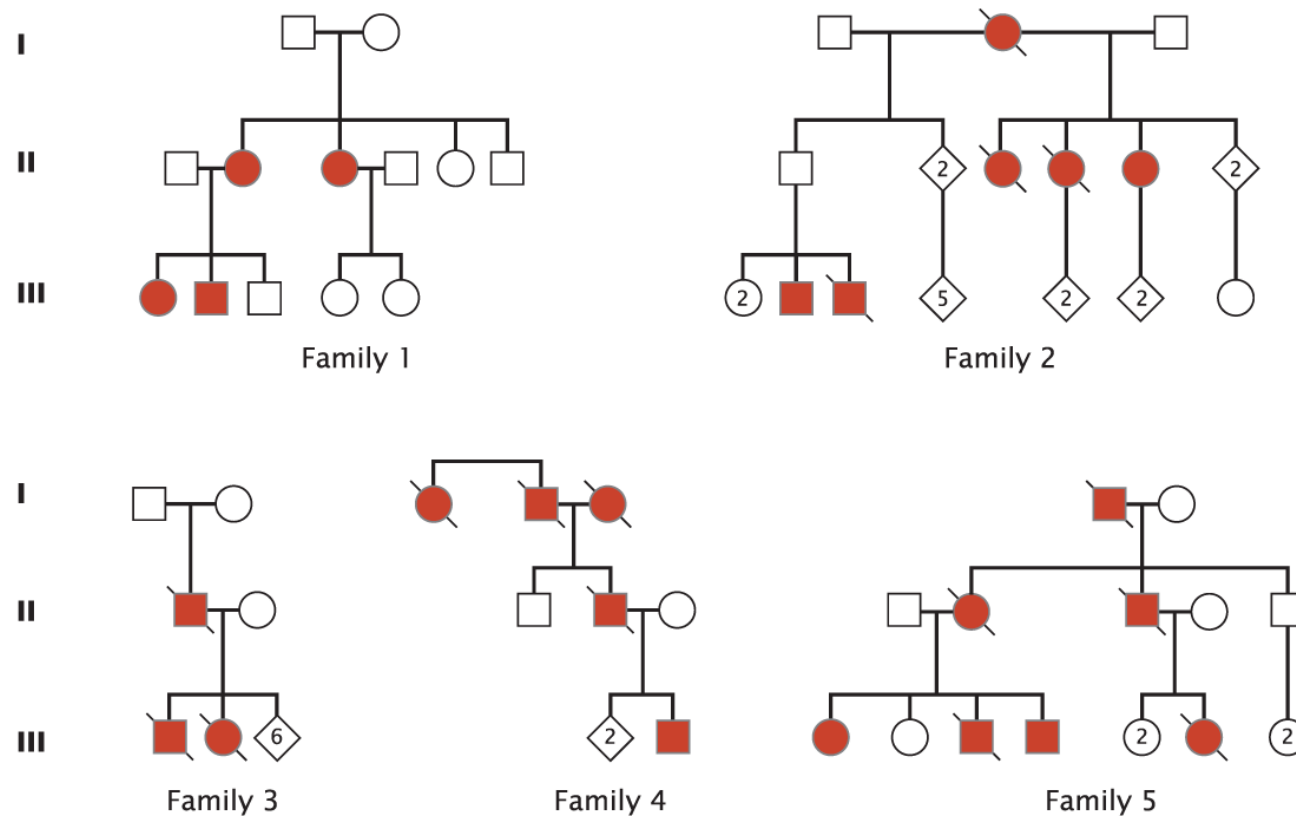


Transformed 3T3 cells



Src oncogene transforms normal cells to become insensitive to contact inhibition

Cancer as a genetic disease



Pierce, *Genetics: A Conceptual Approach*, 7e © 2020 W. H. Freeman and Company

Pierce Figure 23.1

Pedigrees of families with Li-Fraumeni syndrome.

Common inherited cancer syndromes for which genetic testing is available

Hereditary breast cancer and ovarian cancer syndrome

Genes: BRCA1, BRCA2

Li-Fraumeni syndrome

Gene: P53

Familial adenomatous polyposis

Gene: APC

Retinoblastoma

Gene: RB1

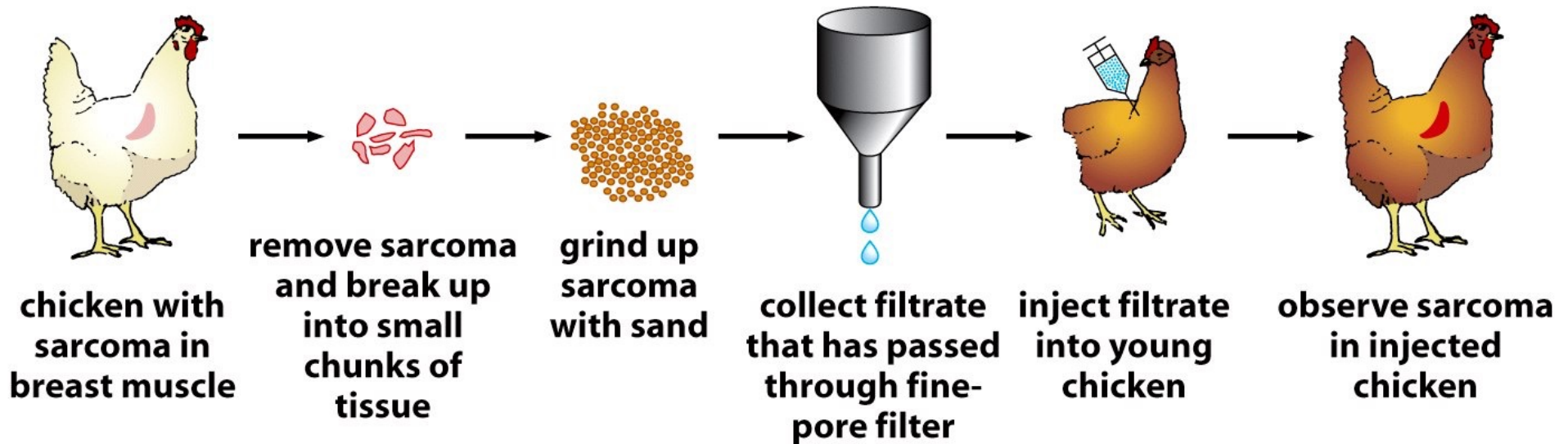
Genes implicated in cancer

Oncogenes = positive regulators driving tumorigenesis

Tumor suppressor genes = negative regulators that are inactivated in cancer

DNA repair genes = prevents mutations maintaining DNA integrity

Tumor virus



The Biology of Cancer (© Garland Science 2007)

1910: Peyton Rous discovered that particles smaller than bacteria could reproducibly induce sarcomas in chickens. This led to discovery of tumor virus (RNA tumor virus).

Virus must carry a cancer-causing gene!!!!

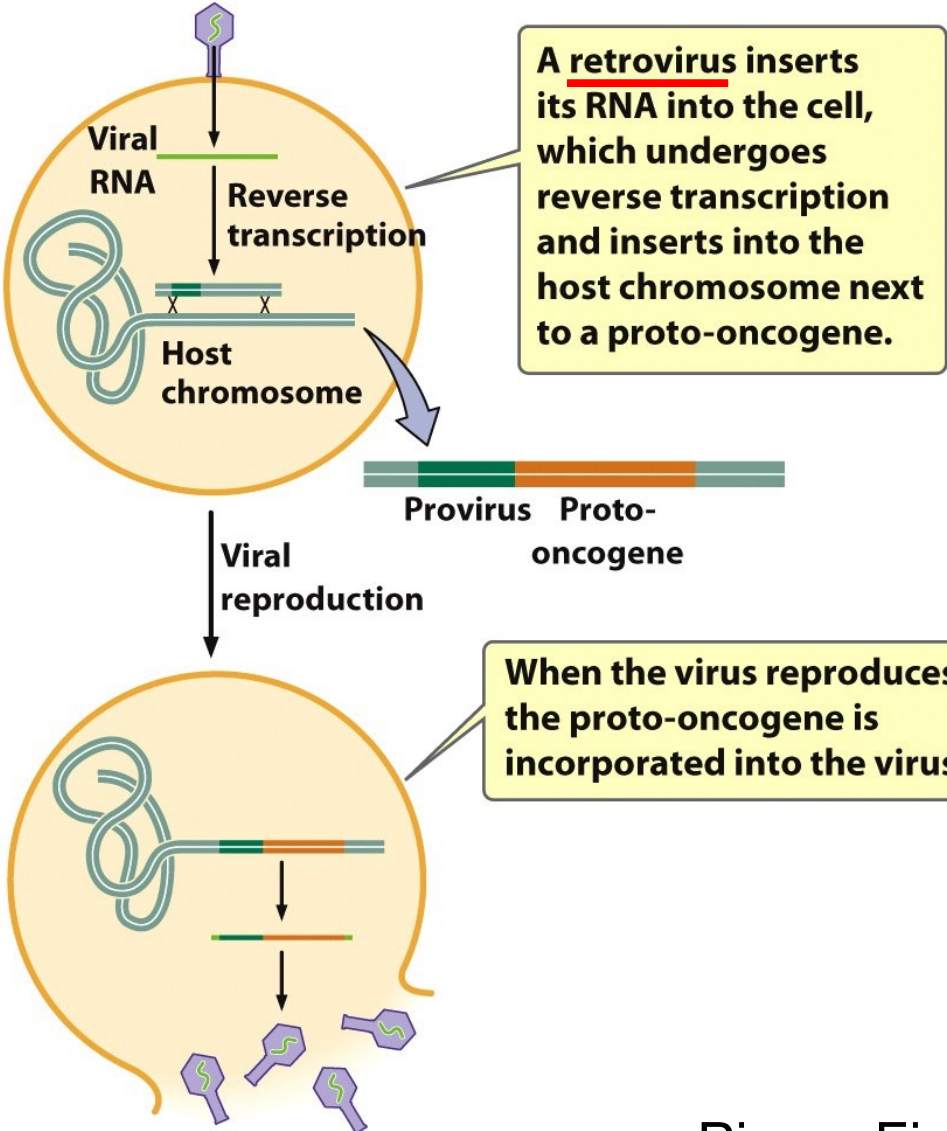
Oncogenes :

First oncogene src (v-Src) was isolated from Rous sarcoma virus in 1970.

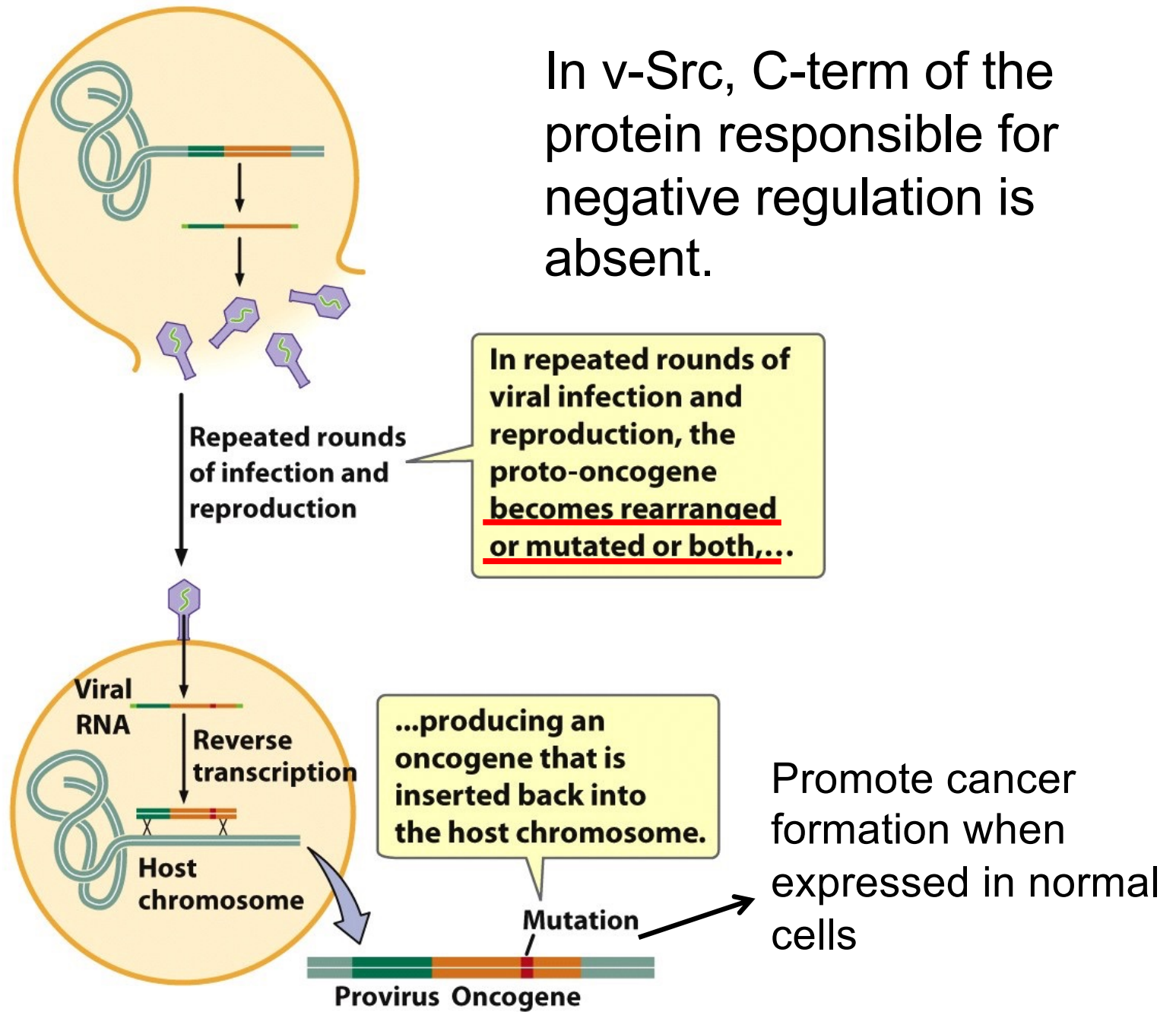
In late 1970s, search for related sequences in normal cells discovered proto-oncogenes (c-Src).

proto-oncogenes are required for normal cellular function, but when mutated, become oncogenes and promote cancer formation

Viral Oncogenes

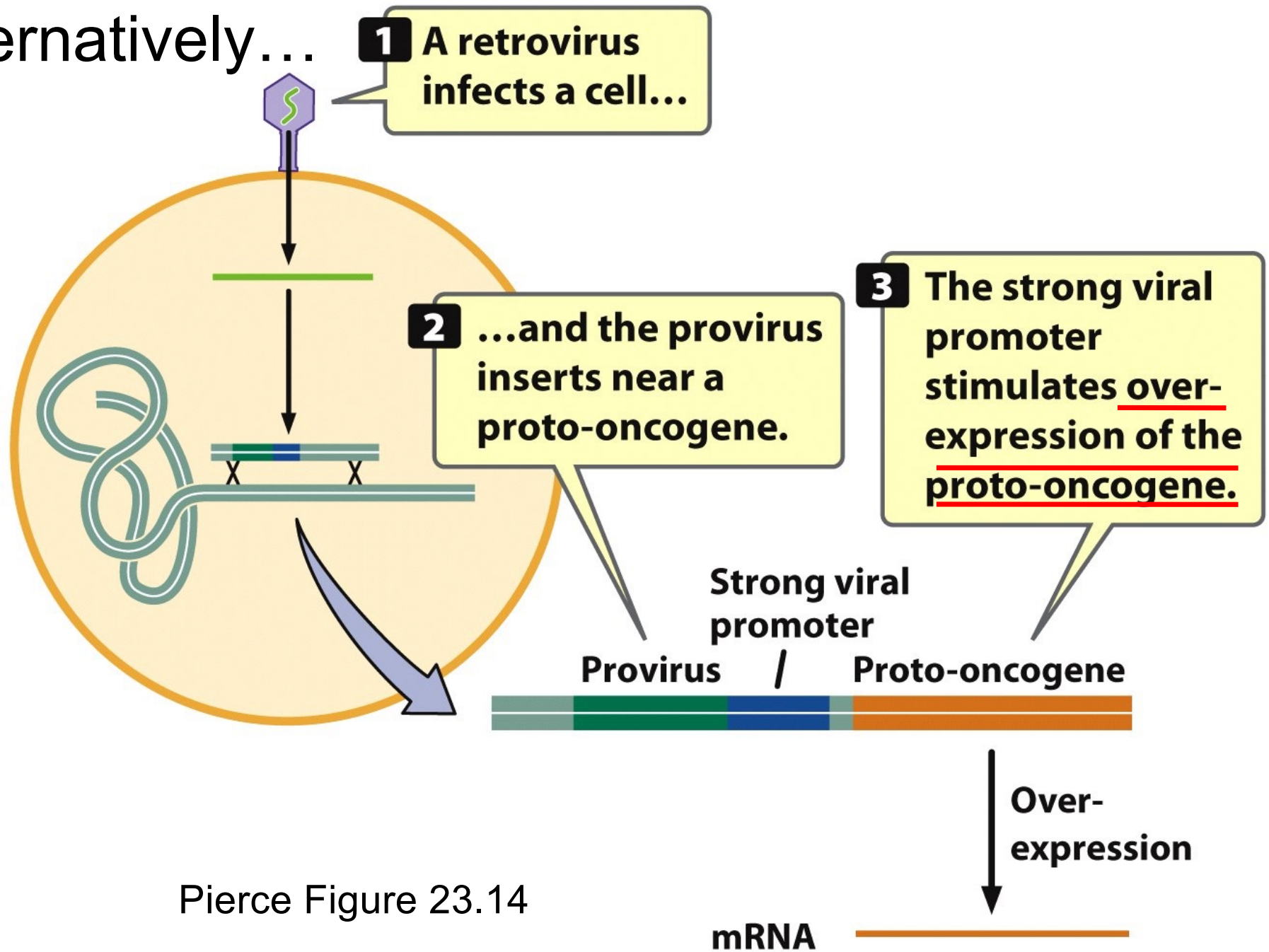


Pierce Figure 23.14



Pierce Figure 23.14

Alternatively...



Pierce Figure 23.14

Non-Viral transformation of proto-oncogenes to oncogenes

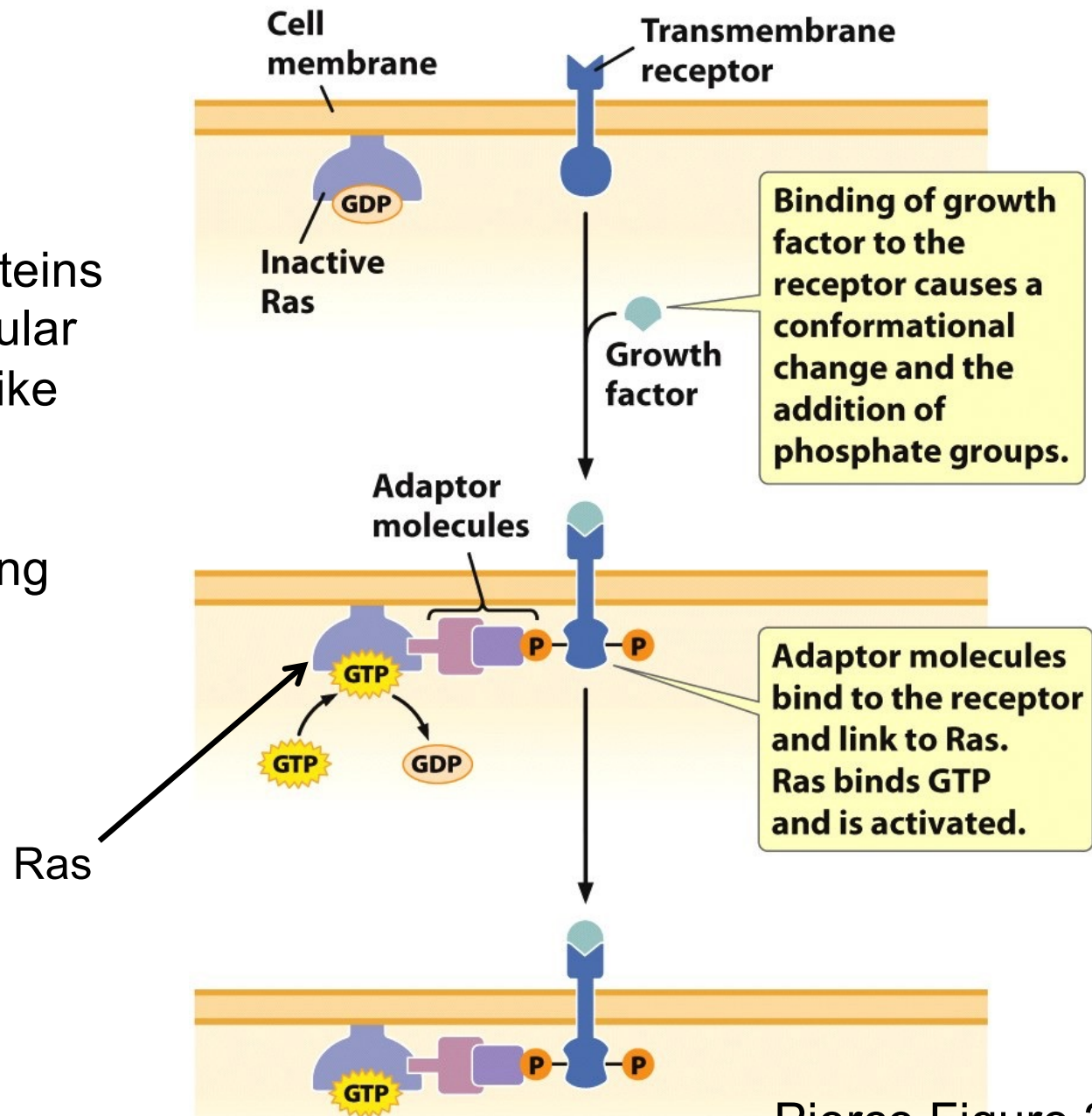
In human, most proto-oncogenes are activated in the absence of viral infection:

- ◆ Mutations in coding sequences
- ◆ Chromosome abnormalities: increased expression, fusion protein (cancer specific form).

Receptor tyrosine kinases and Ras

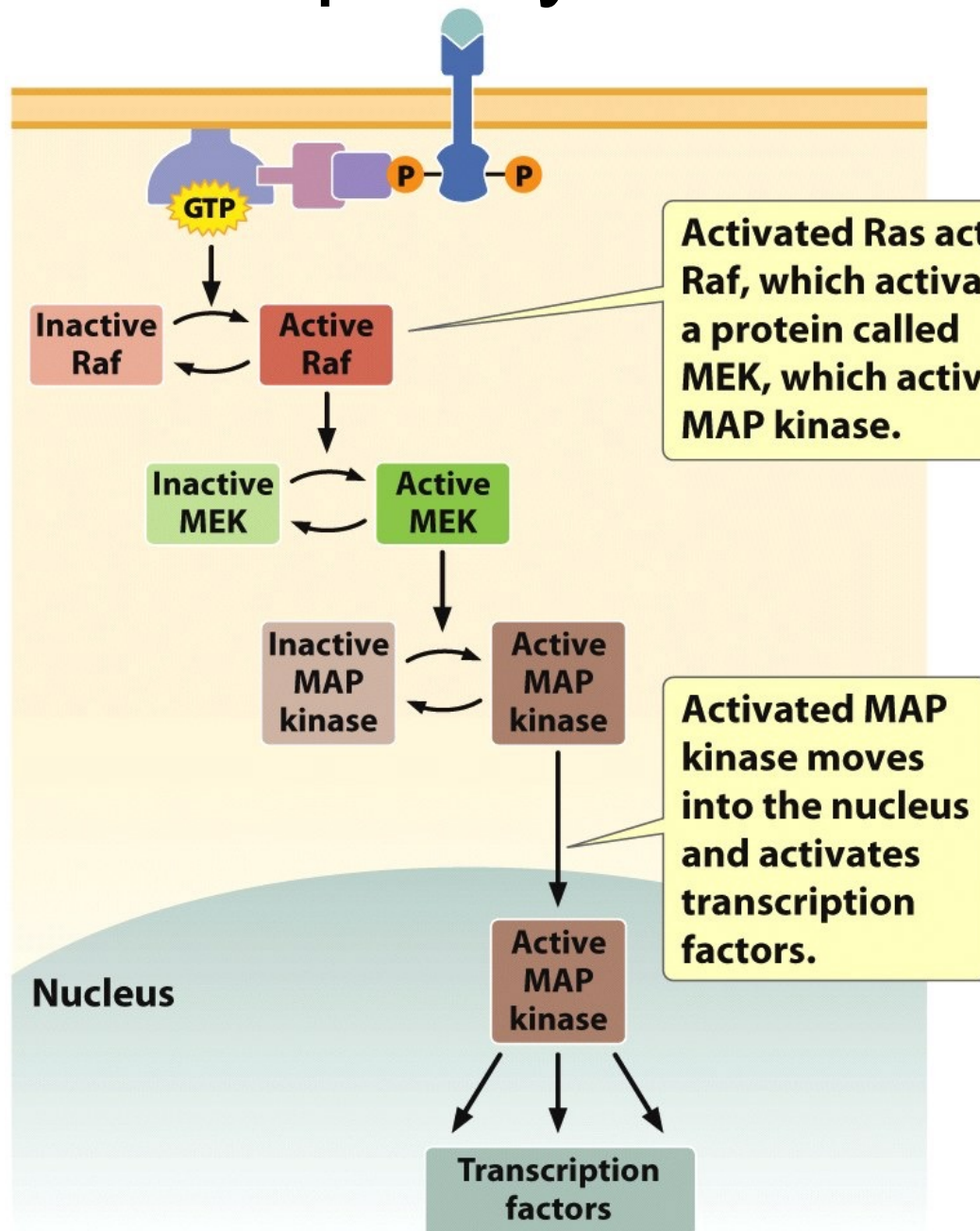
RTK: cell surface proteins that bind to extra cellular signaling molecules like growth factors.

Binding of the signaling molecule result in phosphorylation on tyrosine residues



Pierce Figure 23.9

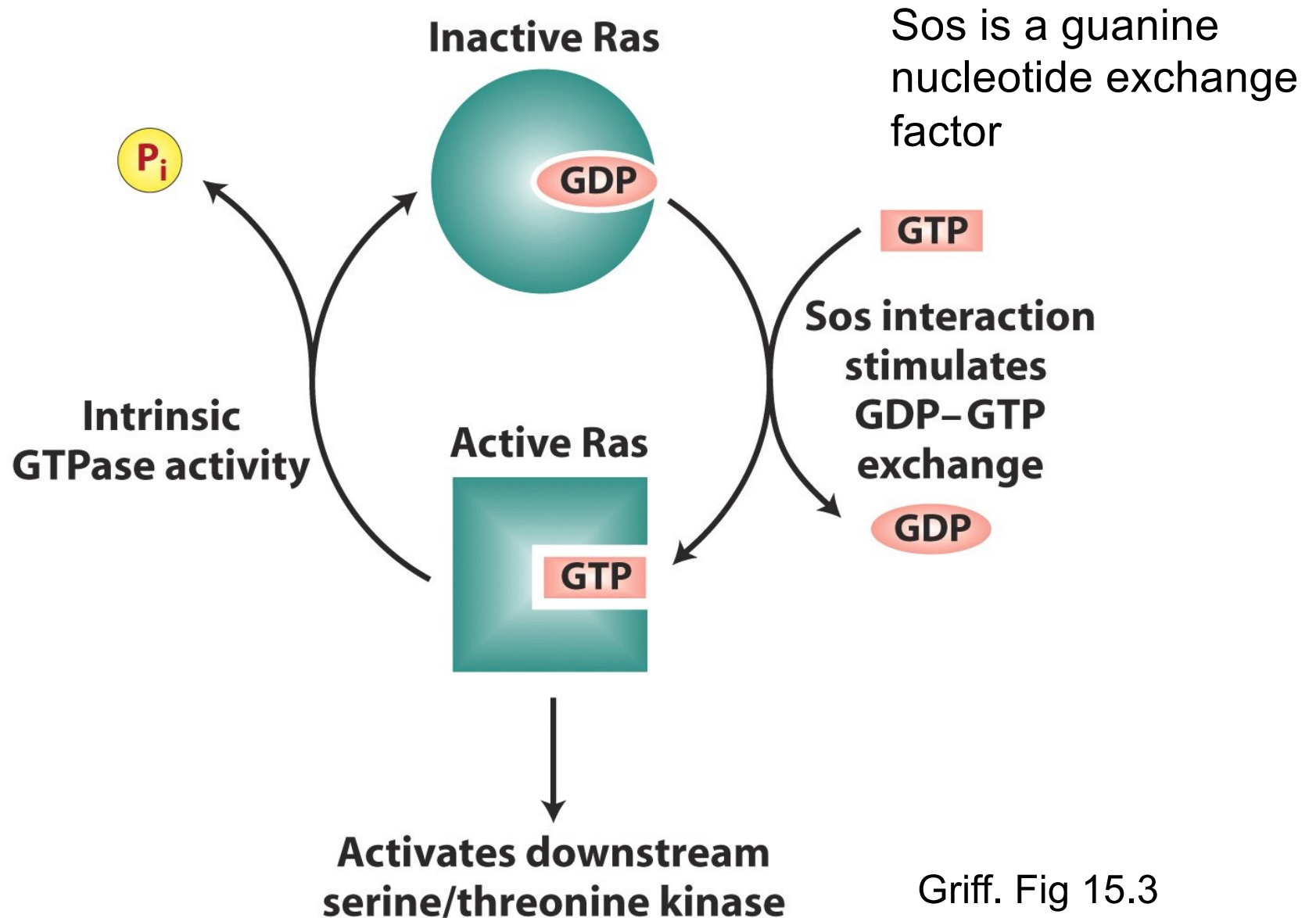
Receptor tyrosine kinases and Ras



GTP bound Ras (not kinase) binds Raf (kinase), which initiates a cascade of phosphorylation events.

Activation of TFs that promote cellular growth

Ras activity is carefully controlled in normal cells

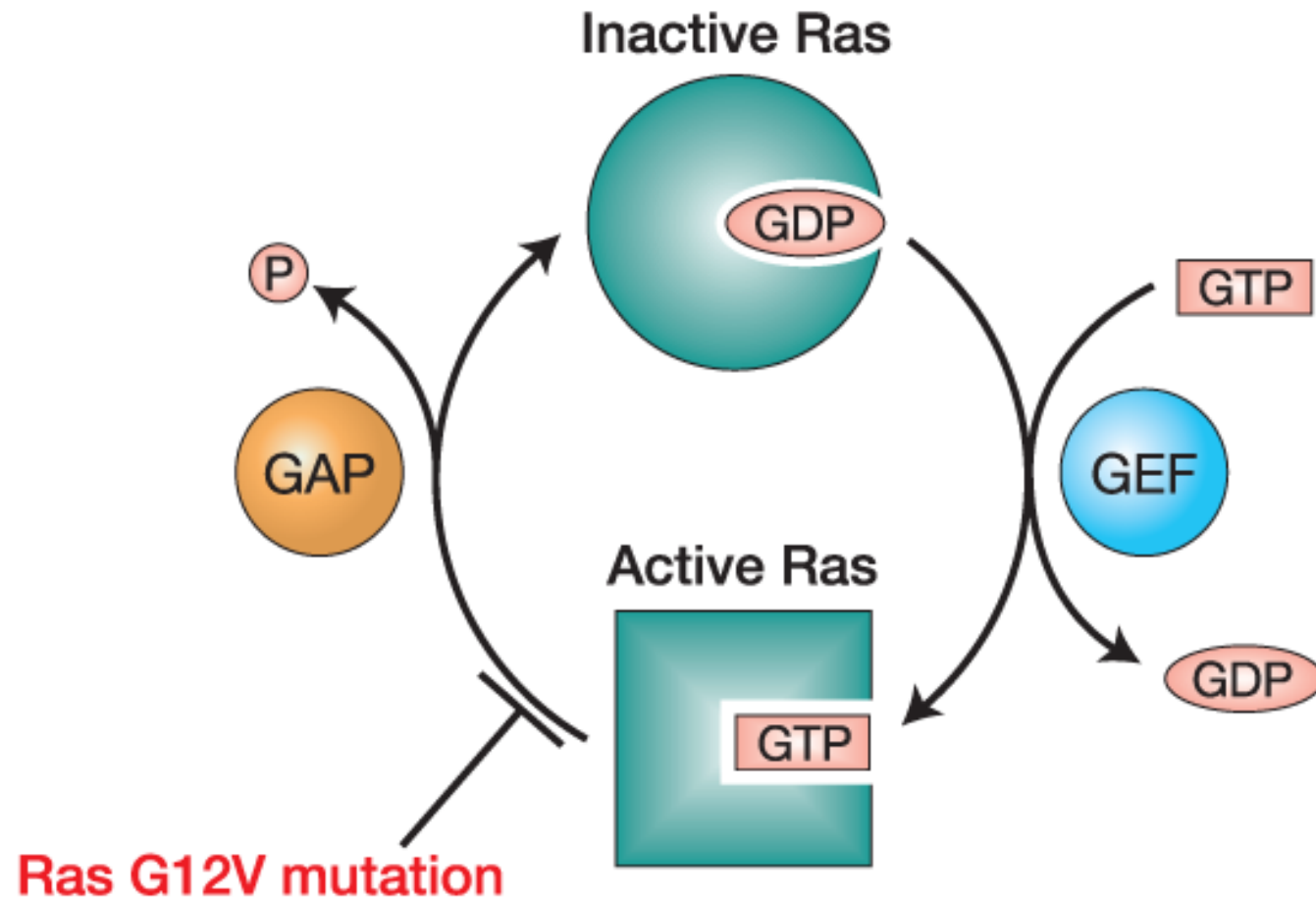


Griff. Fig 15.3

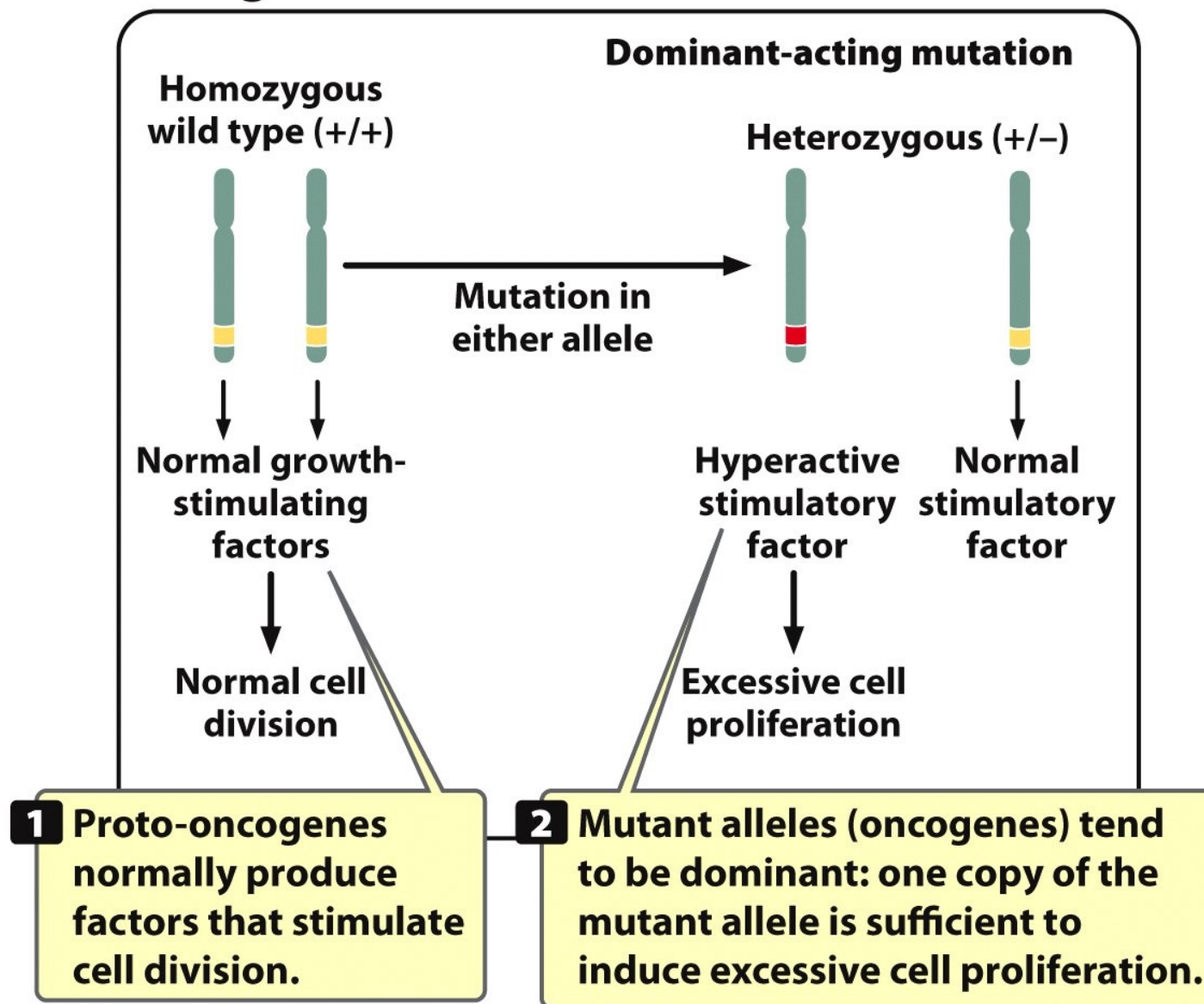
A single point mutation in Ras is sufficient to drive tumorigenesis

	Amino acid 10			Amino acid 15		
	Gly	Ala	Gly	Gly	Val	Gly
<i>ras</i> wild-type DNA	GGC	GCC	GGC	GGT	GTG	GGC
			↓			
<i>ras</i> oncogene DNA	GGC	GCC	GTC	GGT	GTG	GGC
			Val			

Oncogenic activation of Ras



Oncogenes

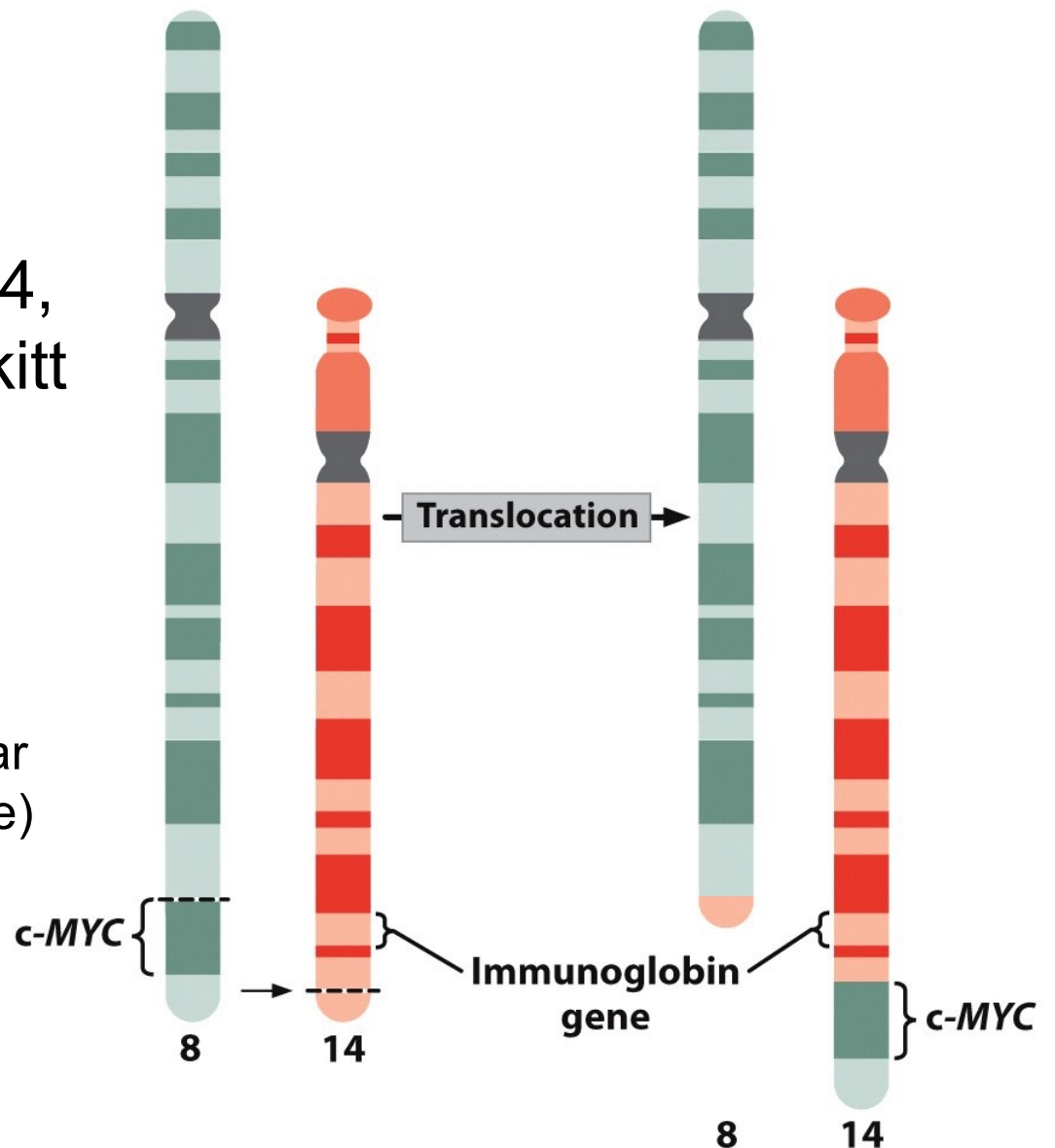


Mutation in a single cell, not in every cells in an organism

Chromosomal changes associated with certain types of cancers

Reciprocal translocation
between Ch 8 and Ch 14,
 $t(8:14)$, observed in Burkitt
lymphoma (B-cell)

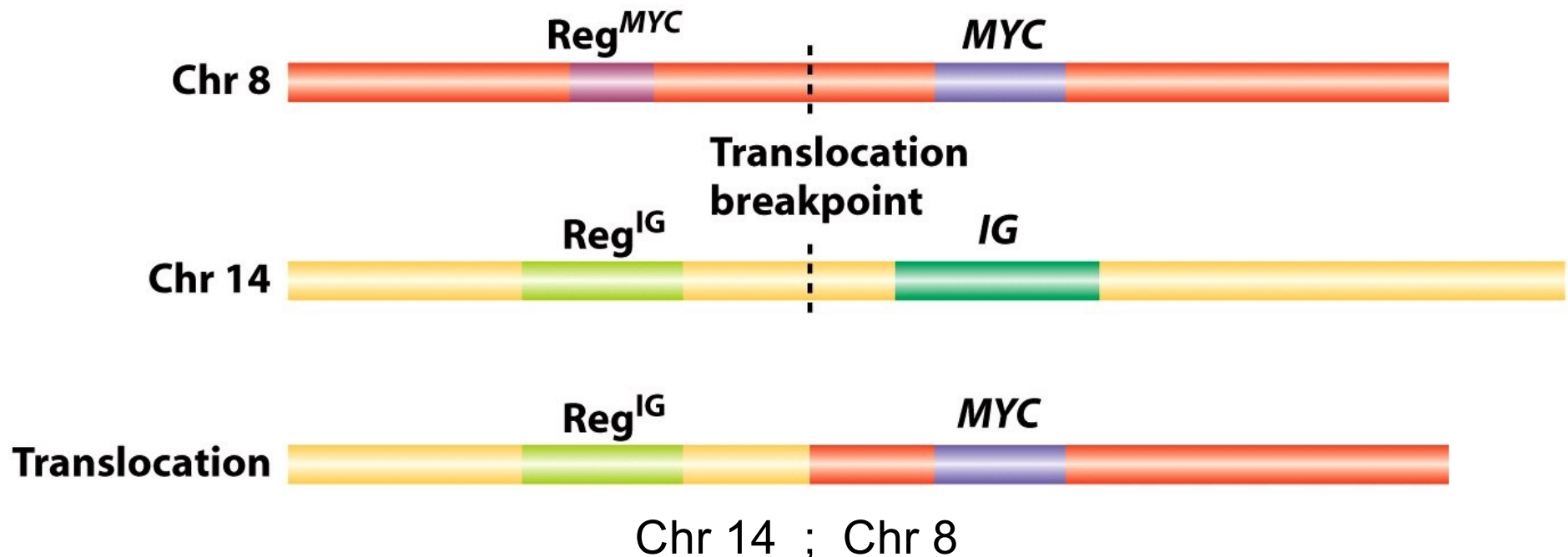
c-Myc: Transcription factor whose
normal function is to promote cellular
growth/proliferation (proto-oncogene)



Pierce Figure 23.12

Cancer by translocation:

Relocation of an oncogene next to a novel regulatory element: Burkitt lymphoma



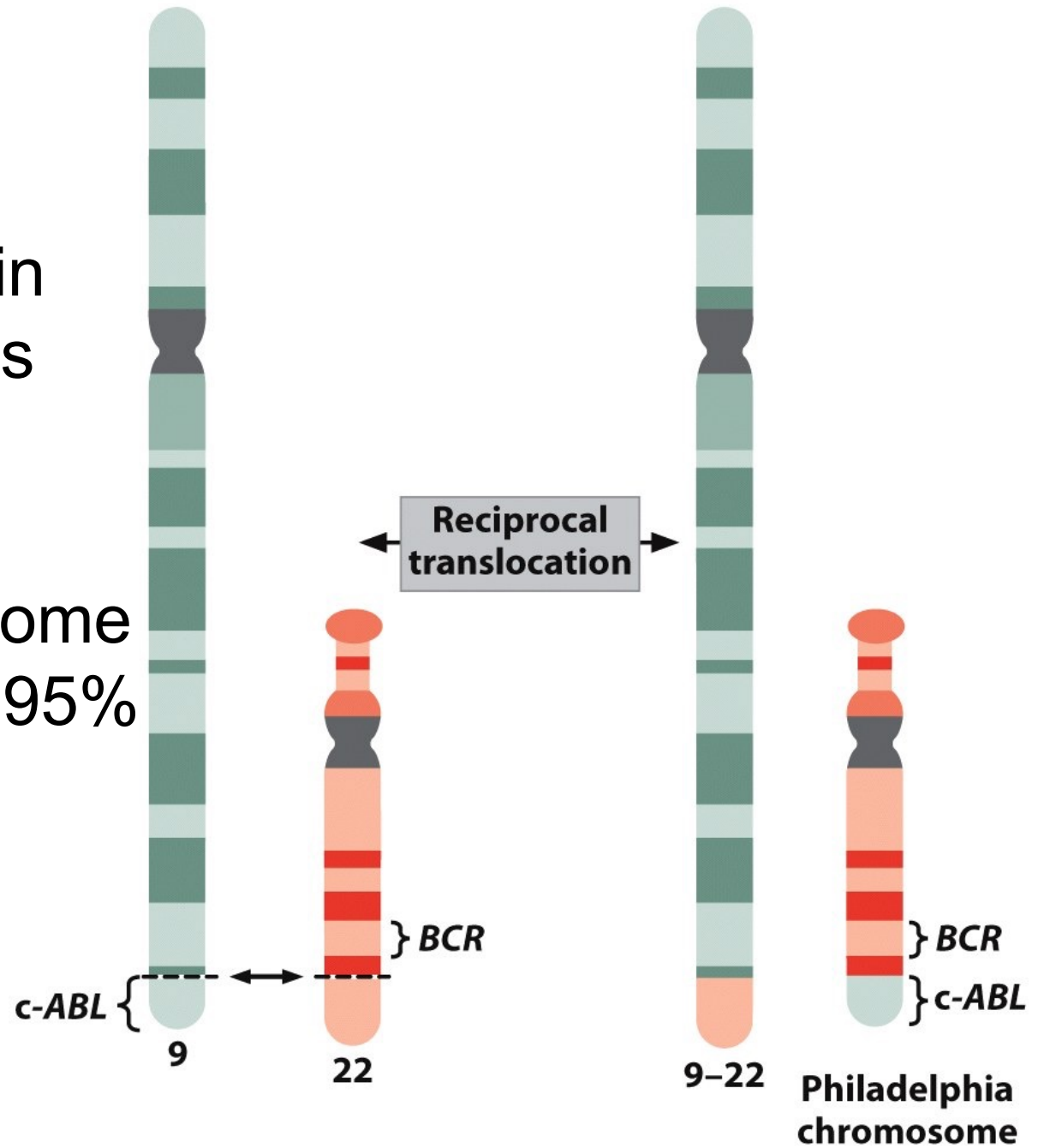
Griff. Fig 17.35a

The coding sequence of Myc is NOT changed. However, the Myc protein is **OVEREXPRESSED** by IG regulatory elements.

Reciprocal
translocation
frequently observed in
Chronic Myelogenous
Leukemia (CML)

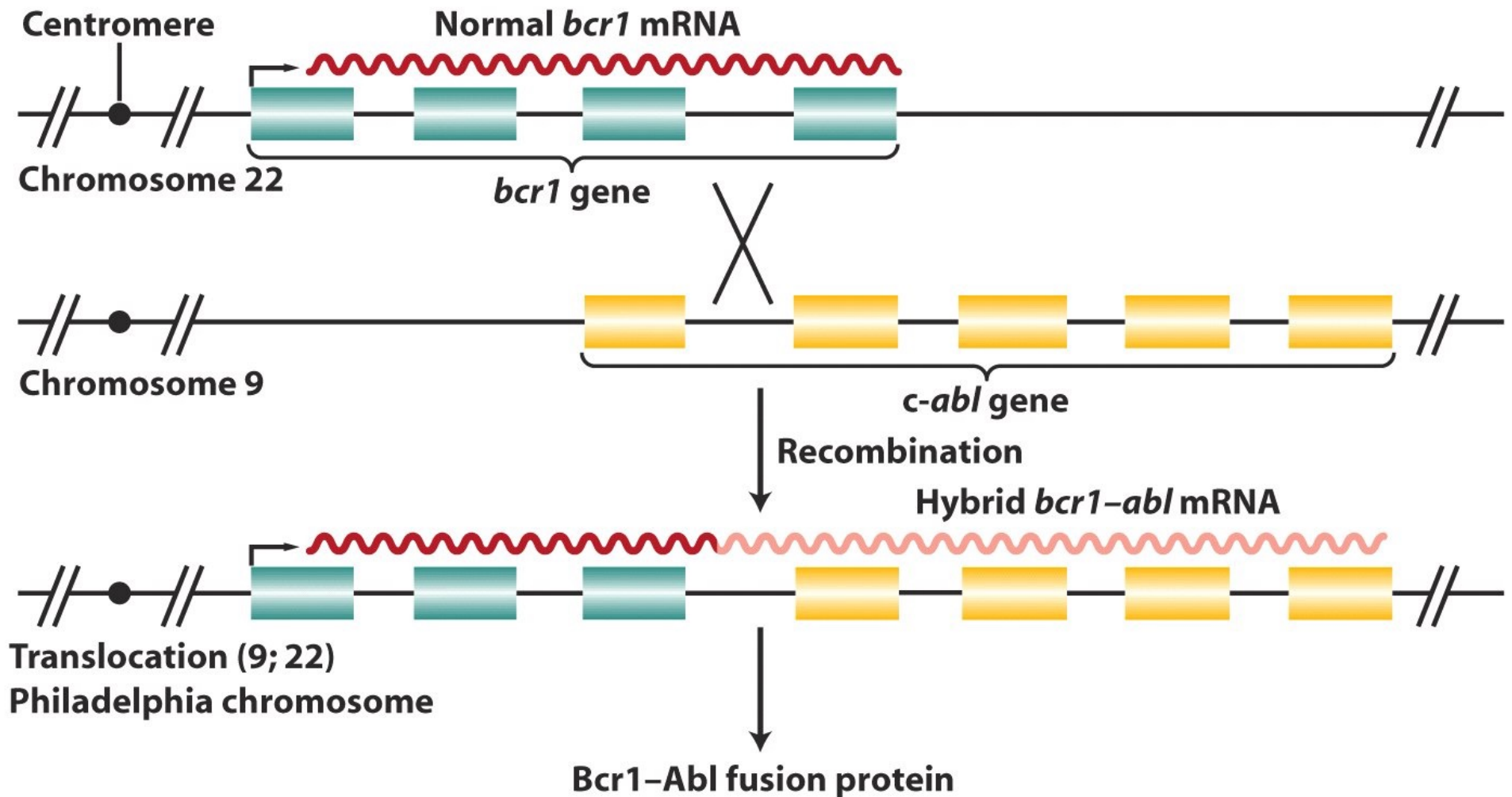
Philadelphia chromosome
 $t(9:22)$ is observed in 95%
of people with CML

c-Abl: protein kinase in
involved in many cellular
processes (proto-oncogene)



Cancer by translocation: A new protein !

... A chimeric protein at the molecular level



Bcr1-Abl becomes a hyperactivated kinase